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Roll No.:- 37

Div:- A

Title:- Implement Greedy search algorithm for selection sort

Program :-

Greedy search Algorithm for selection sort program

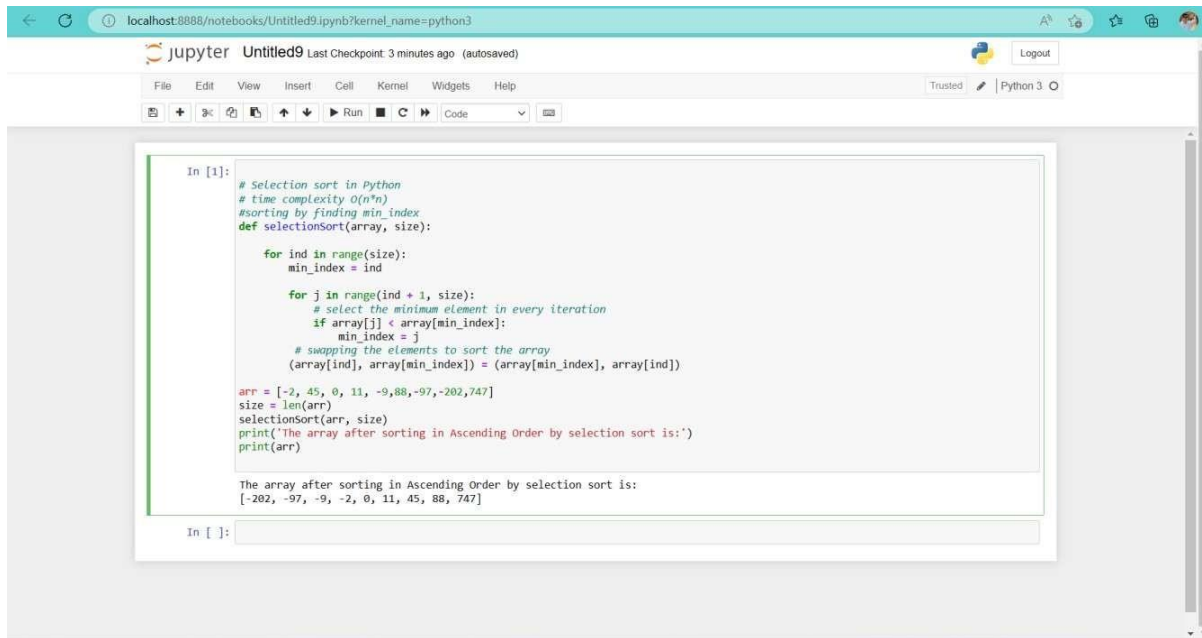
```
# Selection sort in Python
# time complexity O(n*n)
# sorting by finding min_index def
selectionSort(array, size):

    for ind in range(size):
        min_index = ind

        for j in range(ind + 1, size):
            # select the minimum element in every iteration
            if array[j] < array[min_index]:
                min_index = j
            # swapping the elements to sort the array
            (array[ind], array[min_index]) = (array[min_index], array[ind])

arr = [-2, 45, 0, 11, -9, 88, -97, -202, 747] size = len(arr)
selectionSort(arr, size) print('The array after sorting in Ascending
Order by selection sort is:')
```

print(arr)



The image shows a Jupyter Notebook interface in a web browser. The browser address bar shows 'localhost:8888/notebooks/Untitled9.ipynb?kernel_name=python3'. The Jupyter interface includes a top bar with the 'jupyter' logo, the notebook name 'Untitled9', and a 'Last Checkpoint: 3 minutes ago (autosaved)' status. Below this is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Cell', 'Kernel', 'Widgets', and 'Help'. A toolbar contains icons for file operations and a 'Run' button. The main area displays a code cell with the following Python code:

```
In [1]:  
# Selection sort in Python  
# time complexity O(n^2)  
# sorting by finding min_index  
def selectionSort(array, size):  
    for ind in range(size):  
        min_index = ind  
        for j in range(ind + 1, size):  
            # select the minimum element in every iteration  
            if array[j] < array[min_index]:  
                min_index = j  
            # swapping the elements to sort the array  
            (array[ind], array[min_index]) = (array[min_index], array[ind])  
  
arr = [-2, 45, 0, 11, -9, 88, -97, -202, 747]  
size = len(arr)  
selectionSort(arr, size)  
print('The array after sorting in Ascending Order by selection sort is:')  
print(arr)  
  
The array after sorting in Ascending Order by selection sort is:  
[-202, -97, -9, -2, 0, 11, 45, 88, 747]
```

Below the code cell, the prompt 'In []:' is followed by an empty input field.